

USING RESOURCES – WASTE WATER TREATMENT

KEY VOCABULARY

Sewage: waste water containing organic matter and microbes.

Screening: removing large solids and grit.

Sedimentation: settling out solids into sludge and effluent.

Aerobic digestion: bacteria break down effluent using oxygen.

COMMON MISCONCEPTIONS

~~— Sewage treatment is the same as making potable water.~~

✓ It removes harmful matter before release, not for drinking.

~~— All bacteria are removed.~~

✓ Helpful bacteria are USED to break down waste.

~~— Sludge is thrown away as useless.~~

✓ Sludge can be digested to make biogas/fertiliser.

TREATMENT STAGES

Screening — remove grit and large solids.

Sedimentation — solids settle as sludge.

Effluent (liquid) treated by aerobic bacteria.

Sludge treated by anaerobic bacteria → biogas.

Treated water can be safely released.

1 FIRST STEP

State why sewage is **screened** first. [1 mark]

2 SEDIMENTATION

Explain what happens during sedimentation. [2 marks]

3 EFFLUENT

Explain how the liquid effluent is treated. [2 marks]

4 USE OF SLUDGE

Suggest one useful product from digested sludge. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Liquid ____ is treated by ____ bacteria using oxygen.

aerobic

anaerobic

effluent

sludge

6 COMPARE SOURCES

Explain why treating sewage needs more steps than treating ground water. [2 marks]